

Complex Analysis Homework 1

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1. Prove that

$$\int_0^\infty \sin(x^2) dx = \int_0^\infty \cos(x^2) dx = \frac{\sqrt{2\pi}}{4}.$$

These are the Fresnel integrals. Here, $\int_0^\infty$ is interpreted as $\lim_{R \rightarrow \infty} \int_0^R$.

[Hint: Integrate the function e^{-z^2} over the path in Figure 14. Recall that $\int_{-\infty}^\infty e^{-x^2} dx = \sqrt{\pi}$.]

证明 考虑复函数 $f(z) = e^{-z^2}$, 积分路径取第一象限中由以下三段组成的闭合曲线 Γ_R :

1. 实轴上的线段 C_1 : $z = x$, $x \in [0, R]$;
2. 圆弧 C_2 : $z = Re^{i\theta}$, $\theta \in [0, \pi/4]$;
3. 射线 C_3 : $z = re^{i\pi/4}$, $r \in [0, R]$ 。

由柯西积分定理:

$$\int_{C_1} e^{-z^2} dz + \int_{C_2} e^{-z^2} dz - \int_{C_3} e^{-z^2} dz = 0.$$

对于第一段:

$$\int_{C_1} e^{-z^2} dz = \int_0^R e^{-x^2} dx \xrightarrow{R \rightarrow \infty} \frac{\sqrt{\pi}}{2}.$$

对于第二段: 令 $z = Re^{i\theta}$, 则 $z^2 = R^2 e^{i2\theta}$, $dz = iRe^{i\theta} d\theta$,

$$\left| \int_{C_2} e^{-z^2} dz \right| \leq \int_0^{\pi/4} |e^{-R^2(\cos 2\theta + i \sin 2\theta)}| \cdot R d\theta = R \int_0^{\pi/4} e^{-R^2 \cos 2\theta} d\theta.$$

当 $\theta \in [0, \pi/4]$ 时, $\cos 2\theta \geq 1 - 4\theta/\pi$, 因此:

$$R \int_0^{\pi/4} e^{-R^2(1-4\theta/\pi)} d\theta = Re^{-R^2} \int_0^{\pi/4} e^{4R^2\theta/\pi} d\theta = \frac{\pi}{4R} e^{-R^2} (e^{R^2} - 1) \xrightarrow{R \rightarrow \infty} 0.$$

对于第三段：令 $z = re^{i\pi/4}$ ，则 $z^2 = ir^2$ ， $dz = e^{i\pi/4}dr$ ，

$$\int_{C_3} e^{-z^2} dz = e^{i\pi/4} \int_0^R e^{-ir^2} dr = \frac{\sqrt{2}}{2}(1+i) \int_0^R (\cos r^2 - i \sin r^2) dr.$$

令 $R \rightarrow \infty$ ，得：

$$\frac{\sqrt{\pi}}{2} = \frac{\sqrt{2}}{2}(1+i) \left(\int_0^\infty \cos x^2 dx - i \int_0^\infty \sin x^2 dx \right).$$

$$(1-i) \frac{\sqrt{2\pi}}{4} = \int_0^\infty \cos x^2 dx - i \int_0^\infty \sin x^2 dx.$$

对比实部虚部即得证。

2. Show that $\int_0^\infty \frac{\sin x}{x} dx = \frac{\pi}{2}$.

[Hint: The integral equals $\frac{1}{2i} \int_{-\infty}^\infty \frac{e^{ix}-1}{x} dx$. Use the indented semicircle.]

证明 考虑复函数 $f(z) = \frac{e^{iz}}{z}$ ，积分路径取上半平面的挖去原点的半圆 Γ_R^ϵ ($R \rightarrow \infty, \epsilon \rightarrow 0$):

1. 实轴上的线段 C_1 : $x \in [\epsilon, R]$;
2. 上半圆弧 C_2 : $z = Re^{i\theta}$, $\theta \in [0, \pi]$;
3. 实轴上的线段 C_3 : $x \in [-R, -\epsilon]$;
4. 小半圆 C_4 : $z = \epsilon e^{i\theta}$, $\theta \in [\pi, 0]$ 。

由柯西积分定理：

$$\int_{C_1} \frac{e^{iz}}{z} dz + \int_{C_2} \frac{e^{iz}}{z} dz + \int_{C_3} \frac{e^{iz}}{z} dz - \int_{C_4} \frac{e^{iz}}{z} dz = 0.$$

当 $R \rightarrow \infty$ 时：

$$\int_{C_2} \frac{e^{iz}}{z} dz \xrightarrow{R \rightarrow \infty} 0.$$

令 $z = \epsilon e^{i\theta}$ ，则 $dz = i\epsilon e^{i\theta} d\theta$ ，

$$\int_{C_4} \frac{e^{iz}}{z} dz = \int_\pi^0 \frac{e^{i\epsilon e^{i\theta}}}{\epsilon e^{i\theta}} \cdot i\epsilon e^{i\theta} d\theta = i \int_\pi^0 (1 + o(1)) d\theta \xrightarrow{\epsilon \rightarrow 0} i\pi.$$

而

$$\int_{C_1} \frac{e^{iz}}{z} dz + \int_{C_3} \frac{e^{iz}}{z} dz = \int_\epsilon^R \frac{e^{ix}}{x} dx + \int_{-R}^{-\epsilon} \frac{e^{ix}}{x} dx = \int_\epsilon^R \frac{e^{ix} - e^{-ix}}{x} dx = 2i \int_\epsilon^R \frac{\sin x}{x} dx.$$

令 $R \rightarrow \infty, \epsilon \rightarrow 0$ ，得：

$$2i \int_0^\infty \frac{\sin x}{x} dx - i\pi = 0 \implies \int_0^\infty \frac{\sin x}{x} dx = \frac{\pi}{2}.$$

证毕。

7. Suppose $f : \mathbb{D} \rightarrow \mathbb{C}$ is holomorphic. Show that the diameter $d = \sup_{z, w \in \mathbb{D}} |f(z) - f(w)|$ of the image of f satisfies

$$2|f'(0)| \leq d.$$

Moreover, it can be shown that equality holds precisely when f is linear, $f(z) = a_0 + a_1 z$.

Note. In connection with this result, see the relationship between the diameter of a curve and Fourier series described in Problem 1, Chapter 4, Book I.

$$[\text{Hint: } 2f'(0) = \frac{1}{2\pi i} \int_{|\zeta|=r} \frac{f(\zeta) - f(-\zeta)}{\zeta^2} d\zeta \text{ whenever } 0 < r < 1.]$$

证明 对于任意 $0 < r < 1$, 使得:

$$2f'(0) = \frac{1}{2\pi i} \int_{|\zeta|=r} \frac{f(\zeta) - f(-\zeta)}{\zeta^2} d\zeta.$$

令 $\zeta = re^{i\theta}$, 则 $d\zeta = ire^{i\theta} d\theta$, 代入得:

$$2f'(0) = \frac{1}{2\pi} \int_0^{2\pi} \frac{f(re^{i\theta}) - f(-re^{i\theta})}{r^2 e^{i2\theta}} \cdot ire^{i\theta} d\theta = \frac{1}{2\pi r} \int_0^{2\pi} [f(re^{i\theta}) - f(-re^{i\theta})] e^{-i\theta} d\theta.$$

对上述等式取模:

$$2|f'(0)| \leq \frac{1}{2\pi r} \int_0^{2\pi} |f(re^{i\theta}) - f(-re^{i\theta})| d\theta \leq \frac{d}{r}$$

由于对于任意 $0 < r < 1$ 上述不等式都成立, 于是令 $r \rightarrow 1$, 有

$$2|f'(0)| \leq d$$

若等号成立, 则 $r \rightarrow 1$ 时, $[f(re^{i\theta}) - f(-re^{i\theta})]e^{-i\theta} = d$, 即有 $f(\zeta) - f(\zeta) = d\zeta, |\zeta| = 1$,

设 $f(z) = \sum_{n=0}^{\infty} a_n z^n$, 则 $f(\zeta) - f(-\zeta) = 2 \sum_{k=0}^{\infty} a_{2k+1} \zeta^{2k+1} = d\zeta, |\zeta| = 1$, 从而对于 $k \geq 1, a_{2k+1} = 0$.

下证所有偶次项必为零。反设存在 $k \geq 1$ 使 $a_{2k} \neq 0$ 。令 $\theta = \pi + \delta$, 则 $e^{i\theta} = -e^{i\delta}$ 。考虑函数

$$H(z) = f(z) - f(-e^{i\delta} z) = a_1(1 + e^{i\delta})z + a_{2k}(1 - e^{i2k\delta})z^{2k} + O(z^{2k+2}).$$

取 z 在单位圆上使 $a_1(1 + e^{i\delta})z$ 与 $a_{2k}(1 - e^{i2k\delta})z^{2k}$ 辐角相同, 则当 $\delta \rightarrow 0$ 时有

$$|H(z)| \geq |a_1| \cdot |1 + e^{i\delta}| + |a_{2k}| \cdot |1 - e^{i2k\delta}| - O(\delta^2).$$

而 $|1 + e^{i\delta}| = 2 \cos \frac{\delta}{2} = 2(1 - \frac{\delta^2}{8} + o(\delta^2))$, $|1 - e^{i2k\delta}| = 2|\sin k\delta| = 2k\delta + o(\delta)$,
且 $d = 2|a_1|$ 。于是

$$|H(z)| \geq 2|a_1| \left(1 - \frac{\delta^2}{8}\right) + 2k|a_{2k}|\delta - O(\delta^2) = d + 2k|a_{2k}|\delta - \frac{d}{8}\delta^2 - O(\delta^2).$$

$\delta \rightarrow 0$ 时, $2k|a_{2k}|\delta > \frac{d}{8}\delta^2 + O(\delta^2)$, 有 $|H(z)| > d$, 矛盾。故所有 $a_{2k} = 0$ 。

8. If f is a holomorphic function on the strip $-1 < y < 1$, $x \in \mathbb{R}$ with

$$|f(z)| \leq A(1 + |z|)^\eta, \quad \eta \text{ a fixed real number}$$

for all z in that strip, show that for each integer $n \geq 0$ there exists $A_n \geq 0$ so that

$$|f^{(n)}(x)| \leq A_n(1 + |x|)^\eta, \quad \text{for all } x \in \mathbb{R}.$$

[Hint: Use the Cauchy inequalities.]

证明 由柯西不等式, 对任意 $x \in \mathbb{R}$, 取以 x 为中心、半径为 $1/2$ 的圆周 $C(x, 1/2)$, 则:

$$|f^{(n)}(x)| \leq \frac{n!}{(1/2)^n} \cdot \sup_{z \in C(x, 1/2)} |f(z)|.$$

对 $z \in C(x, 1/2)$, 有 $|z| \leq |x| + 1/2 \leq 2(1 + |x|)$, 因此:

$$|f(z)| \leq A(1 + |z|)^\eta \leq A(1 + 2(1 + |x|))^\eta \leq A(3(1 + |x|))^\eta = A \cdot 3^\eta (1 + |x|)^\eta.$$

令 $A_n = n! \cdot 2^n \cdot A \cdot 3^\eta$, 则:

$$|f^{(n)}(x)| \leq n! \cdot 2^n \cdot A \cdot 3^\eta (1 + |x|)^\eta = A_n(1 + |x|)^\eta.$$

9. Let Ω be a bounded open subset of $\mathbb{C}$, and $\varphi : \Omega \rightarrow \Omega$ a holomorphic function. Prove that if there exists a point $z_0 \in \Omega$ such that

$$\varphi(z_0) = z_0 \quad \text{and} \quad \varphi'(z_0) = 1$$

then φ is linear.

[Hint: Why can one assume that $z_0 = 0$? Write $\varphi(z) = z + a_n z^n + O(z^{n+1})$ near 0, and prove that if $\varphi_k = \varphi \circ \cdots \circ \varphi$ (where φ appears k times), then $\varphi_k(z) = z + k a_n z^n + O(z^{n+1})$. Apply the Cauchy inequalities and let $k \rightarrow \infty$ to conclude the proof. Here we use the standard O notation, where $f(z) = O(g(z))$ as $z \rightarrow 0$ means that $|f(z)| \leq C|g(z)|$ for some constant C as $|z| \rightarrow 0$.]

证明 令 $\psi(z) = \varphi(z + z_0) - z_0$, 则 $\psi(0) = 0$, $\psi'(0) = \varphi'(z_0) = 1$, 且 ψ 映有界开集 $\Omega - z_0$ 到自身。只需证明 $\psi(z) = z$, 则 $\varphi(z) = z - z_0 + z_0 = z$ 。

设 $\psi(z) = z + a_n z^n + O(z^{n+1})$ ($n \geq 2$, $a_n \neq 0$), 记 $\psi_k = \psi \circ \psi \circ \cdots \circ \psi$, 用数学归纳法证明:

$$\psi_k(z) = z + k a_n z^n + O(z^{n+1}).$$

基例: $k = 1$, 显然成立;

归纳: 若 $\psi_{k-1}(z) = z + (k-1)a_n z^n + O(z^{n+1})$, 则:

$$\psi_k(z) = \psi(\psi_{k-1}(z)) = \psi_{k-1}(z) + a_n (\psi_{k-1}(z))^n + O((\psi_{k-1}(z))^{n+1}) = z + k a_n z^n + O(z^{n+1}).$$

因 $\Omega - z_0$ 有界, 存在 $M > 0$ 使得 $|\psi_k(z)| \leq M$ 对所有 $z \in \Omega - z_0$ 成立。取以 0 为中心、半径 r 的圆周 $C_r \subset \Omega - z_0$, 由柯西不等式:

$$|k a_n| = \left| \frac{\psi_k^{(n)}(0)}{n!} \right| \leq \frac{n! M}{r^n}.$$

即 $|a_n| \leq \frac{M}{k r^n}$, 令 $k \rightarrow \infty$, 得 $a_n = 0$, 与假设矛盾。

因此 $\psi(z) = z$, 即 $\varphi(z) = z$ 。证毕。

11. Let f be a holomorphic function on the disc D_{R_0} centered at the origin and of radius R_0 .

(a) Prove that whenever $0 < R < R_0$ and $|z| < R$, then

$$f(z) = \frac{1}{2\pi} \int_0^{2\pi} f(Re^{i\varphi}) \operatorname{Re} \left(\frac{Re^{i\varphi} + z}{Re^{i\varphi} - z} \right) d\varphi.$$

(b) Show that

$$\operatorname{Re} \left(\frac{Re^{i\gamma} + r}{Re^{i\gamma} - r} \right) = \frac{R^2 - r^2}{R^2 - 2Rr \cos \gamma + r^2}.$$

[Hint: For the first part, note that if $w = R^2/\bar{z}$, then the integral of $f(\zeta)/(\zeta - w)$ around the circle of radius R centered at the origin is zero. Use this, together with the usual Cauchy integral formula, to deduce the desired identity.]

证明 (a) 由柯西积分公式, $f(z) = \frac{1}{2\pi i} \int_{|\zeta|=R} \frac{f(\zeta)}{\zeta - z} d\zeta$. 令 $w = R^2/\bar{z}$, 则 $|w| > R$, 故 $\int_{|\zeta|=R} \frac{f(\zeta)}{\zeta - w} d\zeta = 0$.

两式相减得:

$$f(z) = \frac{1}{2\pi i} \int_{|\zeta|=R} f(\zeta) \left(\frac{1}{\zeta - z} - \frac{1}{\zeta - w} \right) d\zeta$$

计算得:

$$\begin{aligned} \frac{1}{\zeta - z} - \frac{1}{\zeta - w} &= \frac{1}{\zeta - z} - \frac{\bar{z}}{\bar{z}\zeta - \zeta\bar{z}} \\ &= \frac{\zeta\bar{\zeta} - z\bar{z}}{\zeta(\zeta - z)(\bar{\zeta} - \bar{z})} \end{aligned}$$

而

$$\begin{aligned} \operatorname{Re} \left(\frac{\zeta + z}{\zeta - z} \right) &= \frac{\zeta + z}{\zeta - z} + \frac{\overline{\zeta + z}}{\overline{\zeta - z}} \\ &= \frac{\zeta\bar{\zeta} - z\bar{z}}{(\zeta - z)(\bar{\zeta} - \bar{z})} \end{aligned}$$

于是

$$f(z) = \frac{1}{2\pi i} \int_{|\zeta|=R} f(\zeta) \left(\frac{1}{\zeta - z} - \frac{1}{\zeta - w} \right) d\zeta = \frac{1}{2\pi i} \int_{|\zeta|=R} \frac{1}{\zeta} f(\zeta) \operatorname{Re} \left(\frac{\zeta + z}{\zeta - z} \right) d\zeta$$

令 $\zeta = Re^{i\varphi}$, 则 $d\zeta = iRe^{i\varphi}d\varphi$, 代入得:

$$f(z) = \frac{1}{2\pi} \int_0^{2\pi} f(Re^{i\varphi}) \operatorname{Re} \left(\frac{Re^{i\varphi} + z}{Re^{i\varphi} - w} \right) d\varphi$$

(b) 上一问中得到

$$\operatorname{Re} \left(\frac{\zeta + z}{\zeta - z} \right) = \frac{\zeta\bar{\zeta} - z\bar{z}}{(\zeta - z)(\bar{\zeta} - \bar{z})}$$

代入 $\zeta = Re^{i\gamma}, z = r$ 得

$$\operatorname{Re} \left(\frac{Re^{i\gamma} + r}{Re^{i\gamma} - r} \right) = \frac{R^2 - r^2}{R^2 + r^2 - rR(e^{i\gamma} + e^{-i\gamma})} = \frac{R^2 - r^2}{R^2 - 2Rr \cos \gamma + r^2}$$

12. Let u be a real-valued function defined on the unit disc $\mathbb{D}$. Suppose that u is twice continuously differentiable and harmonic, that is,

$$\Delta u(x, y) = 0$$

for all $(x, y) \in \mathbb{D}$.

(a) Prove that there exists a holomorphic function f on the unit disc such that

$$\operatorname{Re}(f) = u.$$

Also show that the imaginary part of f is uniquely defined up to an additive (real) constant.

[Hint: From the previous chapter we would have $f'(z) = 2\partial u/\partial z$. Therefore, let $g(z) = 2\partial u/\partial z$ and prove that g is holomorphic. Why can one find F with $F' = g$? Prove that $\operatorname{Re}(F)$ differs from u by a real constant.]

(b) Deduce from this result, and from Exercise 11, the Poisson integral representation formula from the Cauchy integral formula: If u is harmonic in the unit disc and continuous on its closure, then if $z = re^{i\theta}$ one has

$$u(z) = \frac{1}{2\pi} \int_0^{2\pi} P_r(\theta - \varphi) u(\varphi) d\varphi$$

where $P_r(\gamma)$ is the Poisson kernel for the unit disc given by

$$P_r(\gamma) = \frac{1 - r^2}{1 - 2r \cos \gamma + r^2}.$$

证明 (a) 令 $g(z) = 2\frac{\partial u}{\partial z} = u_x - iu_y$, 由调和性 $\Delta u = u_{xx} + u_{yy} = 0$ 和 u 二阶微分存在, 故 g 满足柯西-黎曼方程, 故 g 全纯。

因 $\mathbb{D}$ 单连通, 存在全纯函数 F 使得 $F' = g$, 则 $\operatorname{Re}(F') = u_x$, $\operatorname{Im}(F') = -u_y$ 。令 $v = \operatorname{Im}(F)$, 则 $v_x = -\operatorname{Re}(F') = -u_x$, $v_y = \operatorname{Im}(F') = -u_y$, 故 $u - \operatorname{Re}(F)$ 的偏导数均为 0, 即 $u = \operatorname{Re}(F) + C$ (C 为实常数)。

令 $f(z) = F(z) + C$, 则 $\operatorname{Re}(f) = u$ 。若另有全纯函数 f_1 使得 $\operatorname{Re}(f_1) = u$, 则 $f - f_1$ 为纯虚函数, 故为虚常数, 从而虚部唯一 (最多相差一个实常数)。

(b) 由 (a), 存在全纯函数 f 使得 $\operatorname{Re}(f) = u$ 。对 f 应用第 11 题的结果:

$$f(z) = \frac{1}{2\pi} \int_0^{2\pi} f(e^{i\varphi}) \operatorname{Re} \left(\frac{e^{i\varphi} + z}{e^{i\varphi} - z} \right) d\varphi.$$

取实部得:

$$u(z) = \frac{1}{2\pi} \int_0^{2\pi} u(e^{i\varphi}) \cdot \frac{1 - r^2}{1 - 2r \cos(\theta - \varphi) + r^2} d\varphi = \frac{1}{2\pi} \int_0^{2\pi} P_r(\theta - \varphi) u(\varphi) d\varphi.$$

证毕。

13. Suppose f is an analytic function defined everywhere in $\mathbb{C}$ and such that for each $z_0 \in \mathbb{C}$ at least one coefficient in the expansion

$$f(z) = \sum_{n=0}^{\infty} c_n (z - z_0)^n$$

is equal to 0. Prove that f is a polynomial.

[Hint: Use the fact that $c_n n! = f^{(n)}(z_0)$ and use a countability argument.]

证明 由泰勒展开系数公式, $c_n = \frac{f^{(n)}(z_0)}{n!}$, 于是条件转化为对于任意 z_0 , 存在 n_0 使得 $f^{(n_0)}(z_0) = 0$

记 $E_n = \{z \in \mathbb{C} \mid f^{(n)}(z) = 0\}$, 则 $\bigcup_{n=0}^{\infty} E_n = \mathbb{C}$. 因 $f^{(n)}$ 全纯, 故 E_n 为闭集。由贝尔纲定理, $\mathbb{C}$ 作为完备度量空间, 必有某个 E_N 包含非空开集 U 。

在 U 上, $f^{(N)}(z) = 0$, 故 $f^{(N)}$ 在 $\mathbb{C}$ 上恒为 0, 因此 f 为次数不超过 $N - 1$ 的多项式。证毕。

14. Suppose that f is holomorphic in an open set containing the closed unit disc, except for a pole at z_0 on the unit circle. Show that if

$$\sum_{n=0}^{\infty} a_n z^n$$

denotes the power series expansion of f in the open unit disc, then

$$\lim_{n \rightarrow \infty} \frac{a_n}{a_{n+1}} = z_0.$$

证明 由第三章定理 1.2: 若 z_0 是 f 的极点, 则存在唯一正整数 m 和 z_0 邻域内非零全纯函数 $h(z)$, 使得

$$f(z) = (z - z_0)^{-m} h(z).$$

因 f 在包含闭单位圆盘的开集上除 z_0 外全纯, 故 $h(z)$ 可延拓为该开集上的全纯函数, 且 $h(z_0) \neq 0$ 。

对 $|z| < 1$, 展开有:

$$\frac{1}{z - z_0} = -\frac{1}{z_0} \cdot \frac{1}{1 - \frac{z}{z_0}} = -\frac{1}{z_0} \sum_{k=0}^{\infty} \left(\frac{z}{z_0}\right)^k = -\sum_{k=0}^{\infty} \frac{z^k}{z_0^{k+1}}.$$

对其求 $m-1$ 阶导数：

$$\frac{1}{(z-z_0)^m} = \frac{(-1)^m}{(m-1)!} \cdot \frac{d^{m-1}}{dz^{m-1}} \left(\frac{1}{z-z_0} \right) = \sum_{k=0}^{\infty} \binom{k+m-1}{m-1} \cdot \frac{(-1)^m}{z_0^{k+m}} z^k.$$

记 z^k 的系数为 d_k ，即

$$d_k = \binom{k+m-1}{m-1} \frac{(-1)^m}{z_0^{k+m}}.$$

因 $h(z)$ 在包含闭单位圆盘的开集上全纯，其幂级数展开为：

$$h(z) = \sum_{n=0}^{\infty} b_n z^n,$$

且收敛半径 $R > 1$ ，故存在常数 M 使得 $|b_n| \leq MR^{-n}$ 。

于是 $f(z) = (\sum_{k=0}^{\infty} d_k z^k) (\sum_{n=0}^{\infty} b_n z^n) = \sum_{n=0}^{\infty} a_n z^n$ ，其中

$$a_n = \sum_{k=0}^n d_k b_{n-k} = \sum_{j=0}^n d_{n-j} b_j.$$

代入 d_k 的表达式得

$$a_n = \frac{(-1)^m}{z_0^m} \sum_{j=0}^n \binom{n-j+m-1}{m-1} z_0^{-(n-j)} b_j = \frac{(-1)^m}{z_0^{n+m}} \sum_{j=0}^n \binom{n-j+m-1}{m-1} z_0^j b_j.$$

令

$$S_n = \sum_{j=0}^n \binom{n-j+m-1}{m-1} z_0^j b_j.$$

由于 $|b_j| \leq MR^{-j}$ 且 $\binom{n-j+m-1}{m-1} \leq C(n-j)^{m-1} \leq Cn^{m-1}$ ，对任意固定的 N 有

$$\left| \sum_{j=N+1}^n \binom{n-j+m-1}{m-1} z_0^j b_j \right| \leq Cn^{m-1} \sum_{j=N+1}^{\infty} R^{-j} = Cn^{m-1} \frac{R^{-N}}{1-1/R}.$$

取 N 充分大可使此项相对于 n^{m-1} 任意小。而对 $j \leq N$ ，当 $n \rightarrow \infty$ 时，

$$\binom{n-j+m-1}{m-1} = \binom{n+m-1}{m-1} \left(1 + O\left(\frac{1}{n}\right) \right),$$

因此

$$\sum_{j=0}^N \binom{n-j+m-1}{m-1} z_0^j b_j = \binom{n+m-1}{m-1} \sum_{j=0}^N z_0^j b_j + O(n^{m-2}).$$

令 $N \rightarrow \infty$, 得

$$S_n = \binom{n+m-1}{m-1} h(z_0) + o(n^{m-1}),$$

其中 $h(z_0) = \sum_{j=0}^{\infty} b_j z_0^j \neq 0$ 。于是

$$a_n = \frac{(-1)^m}{z_0^{n+m}} \left(\binom{n+m-1}{m-1} h(z_0) + o(n^{m-1}) \right).$$

同理,

$$a_{n+1} = \frac{(-1)^m}{z_0^{n+1+m}} \left(\binom{n+m}{m-1} h(z_0) + o(n^{m-1}) \right).$$

因此

$$\frac{a_n}{a_{n+1}} = z_0 \cdot \frac{\binom{n+m-1}{m-1}}{\binom{n+m}{m-1}} \cdot \frac{1+o(1)}{1+o(1)}.$$

而

$$\frac{\binom{n+m-1}{m-1}}{\binom{n+m}{m-1}} = \frac{(n+m-1)!}{(m-1)!n!} \cdot \frac{(m-1)!(n+1)!}{(n+m)!} = \frac{n+1}{n+m} \rightarrow 1,$$

故 $\lim_{n \rightarrow \infty} \frac{a_n}{a_{n+1}} = z_0$ 。

1. Here are some examples of analytic functions on the unit disc that cannot be extended analytically past the unit circle. The following definition is needed. Let f be a function defined in the unit disc $\mathbb{D}$, with boundary circle C . A point w on C is said to be regular for f if there is an open neighborhood U of w and an analytic function g on U , so that $f = g$ on $\mathbb{D} \cap U$. A function f defined on $\mathbb{D}$ cannot be continued analytically past the unit circle if no point of C is regular for f .

(a) Let

$$f(z) = \sum_{n=0}^{\infty} z^{2^n} \quad \text{for } |z| < 1.$$

Notice that the radius of convergence of the above series is 1. Show that f cannot be continued analytically past the unit disc.

[Hint: Suppose $\theta = 2\pi p/2^k$, where p and k are positive integers. Let $z = re^{i\theta}$; then $|f(re^{i\theta})| \rightarrow \infty$ as $r \rightarrow 1$.]

(b) Fix $0 < \alpha < \infty$. Show that the analytic function f defined by

$$f(z) = \sum_{n=0}^{\infty} 2^{-n\alpha} z^{2^n} \quad \text{for } |z| < 1$$

extends continuously to the unit circle, but cannot be analytically continued past the unit circle.

[Hint: There is a nowhere differentiable function lurking in the background. See Chapter 4 in Book I.]

证明 (a) 取 $\theta = 2\pi p/2^k$ (p, k 为正整数), 令 $z = re^{i\theta}$, 则 $z^{2^n} = r^{2^n} e^{i2^n\theta} = r^{2^n} e^{i2\pi p 2^{n-k}}$ 。

当 $n \geq k$ 时, $e^{i2\pi p 2^{n-k}} = 1$, 故:

$$|f(re^{i\theta})| = \left| \sum_{n=0}^{k-1} r^{2^n} e^{i2^n\theta} + \sum_{n=k}^{\infty} r^{2^n} \right| \geq \sum_{n=k}^{\infty} r^{2^n} - \sum_{n=0}^{k-1} r^{2^n}.$$

令 $r \rightarrow 1^-$, 右侧趋向 ∞ , 故 $f(z)$ 在这些点附近无界, 无法解析延拓到单位圆周外。

(b) 首先, 级数 $\sum_{n=0}^{\infty} 2^{-n\alpha} |z|^{2^n} \leq \sum_{n=0}^{\infty} 2^{-n\alpha} = \frac{1}{1-2^{-\alpha}}$ (收敛), 故 $f(z)$ 连续到单位圆周。

类似 (a), 取 $\theta = 2\pi p/2^k$, 则:

$$|f(re^{i\theta})| \geq \sum_{n=k}^{\infty} 2^{-n\alpha} r^{2^n} - \sum_{n=0}^{k-1} 2^{-n\alpha} r^{2^n}.$$

令 $r \rightarrow 1^-$, 右侧趋向 $\sum_{n=k}^{\infty} 2^{-n\alpha} = \frac{2^{-k\alpha}}{1-2^{-\alpha}} > 0$, 且 $f(z)$ 在这些点的导数无界 (类似无处可微函数), 故无法解析延拓。证毕。